Continuum monitoring ontology

Complete schema atlas

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Companion to the simplified and complete conceptual graphs. This atlas keeps every declared class and property readable at normal print size. It describes the asserted TBox, including the benchmark deployment extension and external vocabulary declarations, not individual users or experimental events.

Schema inventory	Count
Named classes	148
Anonymous class expressions	6
Object properties	167
Datatype properties	99
Annotation properties	14
Subclass axioms	70

Reading the diagrams

Solid arrows are object-property domain-to-range projections, not mandatory links, cardinalities or a workflow. Hollow arrows point from a subclass to its parent. OR identifies an explicit union; multiple asserted domains or ranges are combined with AND. "Not declared" means the source supplies no constraint; no class is invented to fill that gap. Datatype properties are attributes. Annotation properties and additional schema axioms are listed separately.

The simplified graph is a deliberately selected subset. The full map is a zoomable vector supplement, not a figure to shrink onto one paper column. Colors identify modules; names, borders and line styles retain meaning in grayscale. Class/individual punning is marked explicitly. SPARQL and SHACL enforcement is not inferred from a diagram.

Foundation

5 named classes | 0 properties

Classes and asserted parents

Class	subClassOf	English label
mc:MAPESymptom	-	MAPE-K Symptom
mc:NonUser	foaf:Person	Non-User
mc:ProcessingLevel	-	Processing Level
mc:TemporalEntity	-	Temporal Entity
mc:User	foaf:Person	User

Shared contracts

13 named classes | 0 properties

Classes and asserted parents

Class	subClassOf	English label
mc:AIModel	-	AI Model
mc:Artifact	-	Artifact
mc:EvaluationState	mc:State	Evaluation State
mc:FederatedLearningSession	-	Federated Learning Session
mc:ModelTier	-	Model Tier
mc:PayloadType	-	Payload Type
mc:ProcessingPurpose	-	Processing Purpose
mc:RecoveryCondition	-	Recovery Condition
mc:Service	-	Service
mc:ServiceState	mc:State	Service State
mc:State	mc:TemporalEntity	State
mc:TransferEvent	mc:TemporalEntity	Transfer Event
mc:ValidationCampaign	mc:Artifact	Validation Campaign

Topology

16 named classes | 15 properties

Classes and asserted parents

Class	subClassOf	English label
mc:CityArea	mc:UrbanEnvironment	City Area
mc:CloudNode	mc:ComputationalNode	Cloud Node
mc:Community	-	Community
mc:ComputationalNode	saref:Device	Computational Node
mc:DistanceLevel	-	Distance Level
mc:EdgeNode	mc:ComputationalNode	Edge Node
mc:FogNode	mc:ComputationalNode	Fog Node
mc:MistNode	mc:ComputationalNode	Mist Node
mc:MobileDevice	mc:MistNode	Mobile Device
mc:NodeUserRelation	mc:TemporalEntity	Node-User Relation
mc:PopulationDensity	-	Population Density
mc:RestrictedZone [punned]	mc:CityArea	Restricted Zone
mc:RuralZone [punned]	mc:CityArea	Rural Zone
mc:UrbanEnvironment	-	Urban Environment
mc:UrbanZone [punned]	mc:CityArea	Urban Zone
mc:ZoneType	-	Zone type

Object properties

Property	Asserted domain	Asserted range
mc:belongsToCommunity	mc:User	mc:Community
mc:connectsTo	saref:Device	mc:ComputationalNode
mc:coversArea	mc:ComputationalNode	mc:CityArea
mc:hasDistance	mc:NodeUserRelation	mc:DistanceLevel
mc:hasGeometry	mc:CityArea	geo:Geometry
mc:hasMobileDevice	mc:User	mc:MobileDevice
mc:hasNeighborNode	mc:ComputationalNode	mc:ComputationalNode
mc:hasPopulationDensity	mc:CityArea	mc:PopulationDensity
mc:locatedIn	not declared	not declared
mc:locatedInZone	mc:User	mc:CityArea
mc:relatesNode	mc:NodeUserRelation	mc:ComputationalNode
mc:relatesUser	mc:NodeUserRelation	mc:User

Datatype properties

Property	Asserted domain	Asserted range
mc:continuumLevel	mc:ModelTier	xsd:integer
mc:hasElasticity	mc:ComputationalNode	xsd:boolean
mc:isPreferredNode	mc:NodeUserRelation	xsd:boolean

Observability

18 named classes | 35 properties

Classes and asserted parents

Class	subClassOf	English label
mc:AvailabilityLevel	-	Availability Level
mc:BatteryLevel	-	Battery Level
mc:CommunicationLevel	-	Communication Level
mc:ComplianceMetric	-	Compliance Metric
mc:DataContext	mc:TemporalEntity	Data Context
mc:DeviceConnectionStatus	-	Device Connection Status
mc:DeviceState	mc:State	Device State
mc:EnergyLevel	-	Energy Level
mc:MobilityLevel	-	Mobility Level
mc:NodeState	mc:State	Node State
mc:OperationalStatus	-	Operational Status
mc:PerformanceLevel	-	Performance Level
mc:ResidualCapacity	-	Residual Capacity
mc:TrafficWindow	-	Traffic Window
mc:TrustAssessment	mc:TemporalEntity	Trust Assessment
mc:TrustEvidence	-	Trust Evidence
mc:UserState	mc:State	User State
mc:WorkloadLevel	-	Workload Level

Object properties

Property	Asserted domain	Asserted range
mc:contextDeviceState	mc:DataContext	mc:DeviceState
mc:contextNodeState	mc:DataContext	mc:NodeState
mc:contextProcessingLevel	mc:DataContext	mc:ProcessingLevel
mc:contextPurpose	mc:DataContext	mc:ProcessingPurpose
mc:contextZone	mc:DataContext	mc:CityArea
mc:hasAvailability	mc:NodeState	mc:AvailabilityLevel
mc:hasBatteryLevel	mc:DeviceState	mc:BatteryLevel
mc:hasCommunication	mc:NodeState	mc:CommunicationLevel
mc:hasComplianceMetric	mc:ValidationCampaign	mc:ComplianceMetric
mc:hasConnectionStatus	mc:DeviceState	mc:DeviceConnectionStatus
mc:hasDataContext	not declared	mc:DataContext
mc:hasEnergyLevel	mc:NodeState	mc:EnergyLevel
mc:hasMobility	mc:UserState	mc:MobilityLevel
mc:hasNodeState	mc:ComputationalNode	mc:NodeState
mc:hasOperationalStatus	mc:NodeState	mc:OperationalStatus
mc:hasPerformance	mc:NodeState	mc:PerformanceLevel
mc:hasResidualCapacity	mc:NodeState	mc:ResidualCapacity
mc:hasTrustAssessment	mc:NodeState	mc:TrustAssessment

Property	Asserted domain	Asserted range
mc:hasTrustEvidence	mc:TrustAssessment	mc:TrustEvidence
mc:hasUserState	mc:User	mc:UserState
mc:hasWorkload	mc:NodeState	mc:WorkloadLevel
mc:trustAssessmentForState	mc:TrustAssessment	mc:NodeState

Datatype properties

Property	Asserted domain	Asserted range
mc:evidenceContribution	mc:TrustEvidence	xsd:decimal
mc:evidenceFactor	mc:TrustEvidence	xsd:string
mc:hasTrustScore	not declared	xsd:decimal
mc:parametrizedDataReady	mc:DeviceState	xsd:boolean
mc:queuedRequests	mc:NodeState	xsd:integer
mc:resourceUsagePercent	mc:NodeState	xsd:decimal
mc:stateDuration	mc:State	xsd:duration
mc:trustRuleVersion	mc:TrustAssessment	xsd:string
mc:trustWindowEnd	mc:TrustAssessment	xsd:dateTime
mc:trustWindowStart	mc:TrustAssessment	xsd:dateTime
mc:userFeedbackScore	mc:UserState	xsd:integer
mc:validFrom	mc:TemporalEntity	xsd:dateTime
mc:validTo	mc:TemporalEntity	xsd:dateTime

Governance

41 named classes | 93 properties

Classes and asserted parents

Class	subClassOf	English label
mc:AcceptanceProfile	mc:Artifact	Acceptance Profile
mc:AnonymousIdentifier	mc:Identifier	Anonymous Identifier
mc:AuthorizationDecision	mc:TemporalEntity	Authorization Decision
mc:AuthorizationOutcome	-	Authorization Outcome
mc:ConflictResolutionStrategy	-	Conflict Resolution Strategy
mc:ConsentRange	-	Consent Range
mc:ConsentRecord	mc:TemporalEntity	Consent Record
mc:DataCategory	-	Data Category
mc:DataCriticality	-	Data Criticality
mc:DataSensitivity	-	Data Sensitivity
mc:DifferentialPrivacyMechanism	mc:PrivacyMechanism	Differential Privacy Mechanism
mc:DirectIdentifier	mc:Identifier	Direct Identifier
mc:EncryptionMechanism	mc:SecurityMechanism	Encryption Mechanism
mc:FunctionalRequirement	mc:Requirement	Functional Requirement
mc:Identifier	-	Identifier
mc:MechanismSpecification	-	Mechanism Specification
mc:NonFunctionalRequirement	mc:Requirement	Non-functional Requirement
mc:OntologyArtifact	mc:Artifact	Ontology Artifact
mc:Permission	-	Permission
mc:Policy	-	Policy
mc:PolicyArtifact	mc:Artifact	Policy Artifact
mc:PolicyCategory	-	Policy Category
mc:PolicyCategoryRelation	-	Policy Category Relation
mc:PolicyRelationType	-	Policy Relation Type
mc:PolicyType	-	Policy Type
mc:PrivacyMechanism	-	Privacy Mechanism
mc:ProcessingScope	-	Processing Scope
mc:PseudonymousIdentifier	mc:Identifier	Pseudonymous Identifier
mc:QueryCatalog	mc:Artifact	Query Catalog
mc:QuerySpecification	-	Query Specification
mc:QueryType	-	Query Type
mc:Requirement	-	Requirement
mc:RequirementsArtifact	mc:Artifact	Requirements Artifact
mc:Role	-	Access Role
mc:Scenario	-	Scenario
mc:ScenarioArtifact	mc:Artifact	Scenario Artifact
mc:SecurityMechanism	-	Security Mechanism
mc:SemanticContract	mc:TemporalEntity	Semantic Contract
mc:StandardSpecification	-	Standard Specification
mc:ValidationRequirement	mc:Requirement	Validation / Reproducibility Requirement

Class	subClassOf	English label
mc:ValidationResult	-	Validation Result

Object properties

Property	Asserted domain	Asserted range
mc:appliedMechanism	mc:EvaluationState	mc:MechanismSpecification
mc:appliedPolicy	mc:EvaluationState	mc:Policy
mc:appliedSecurityMechanism	not declared	mc:SecurityMechanism
mc:appliesInZone	mc:Policy	mc:CityArea
mc:appliesTo	mc:Permission	not declared
mc:appliesToZoneType	mc:Policy	mc:ZoneType
mc:basedOnConsentRecord	mc:AuthorizationDecision	mc:ConsentRecord
mc:basedOnContract	mc:AuthorizationDecision	mc:SemanticContract
mc:basedOnZone	mc:AuthorizationDecision	mc:CityArea
mc:belongsToPolicyCategory	mc:Policy	mc:PolicyCategory
mc:belongsToQueryCatalog	mc:QuerySpecification	mc:QueryCatalog
mc:consentSubject	mc:ConsentRecord	mc:User
mc:contractSubject	mc:SemanticContract	mc:User
mc:definedInArtifact	not declared	mc:Artifact
mc:governedBy	not declared	mc:Policy
mc:hasActiveConsentRange	mc:User	mc:ConsentRange
mc:hasAuthorizationDecision	not declared	mc:AuthorizationDecision
mc:hasAuthorizationOutcome	mc:AuthorizationDecision	mc:AuthorizationOutcome
mc:hasAuthorizedDataCategory	mc:ConsentRecord	mc:DataCategory
mc:hasConsentRange	(mc:User OR mc:SemanticContract OR mc:Policy OR mc:FederatedLearningSession OR mc:EvaluationState OR mc:ConsentRecord)	mc:ConsentRange
mc:hasConsentRecord	mc:User	mc:ConsentRecord
mc:hasDataCategory	not declared	mc:DataCategory
mc:hasDataCriticality	not declared	mc:DataCriticality
mc:hasDataSensitivity	not declared	mc:DataSensitivity
mc:hasEffectiveConsentRange	mc:AuthorizationDecision	mc:ConsentRange
mc:hasIdentifier	mc:User	mc:Identifier
mc:hasPermission	mc:Role	mc:Permission
mc:hasPolicyRelationType	mc:PolicyCategoryRelation	mc:PolicyRelationType
mc:hasPolicyType	mc:Policy	mc:PolicyType
mc:hasProcessingPurpose	(mc:SemanticContract OR mc:Policy OR mc:FederatedLearningSession OR mc:EvaluationState OR mc:Permission OR mc:ConsentRecord OR mc:Service)	mc:ProcessingPurpose
mc:hasProcessingScope	mc:ModelTier	mc:ProcessingScope
mc:hasQueryType	mc:QuerySpecification	mc:QueryType
mc:hasRole	foaf:Person	mc:Role
mc:hasSemanticContract	mc:User	mc:SemanticContract
mc:hasValidationResult	mc:ValidationCampaign	mc:ValidationResult
mc:policyArtifactUsed	mc:EvaluationState	mc:PolicyArtifact

Property	Asserted domain	Asserted range
mc:recommendedMechanism	mc:Policy	mc:MechanismSpecification
mc:relatedRequirement	mc:Policy	mc:Requirement
mc:relationSourceCategory	mc:PolicyCategoryRelation	mc:PolicyCategory
mc:relationTargetCategory	mc:PolicyCategoryRelation	mc:PolicyCategory
mc:requirementsArtifactUsed	mc:EvaluationState	mc:RequirementsArtifact
mc:requiresConsentRange	(mc:Permission OR mc:Policy OR mc:Service OR mc:ServiceState OR mc:AIModel OR mc:FederatedLearningSession OR mc:EvaluationState OR mc:ModelTier)	mc:ConsentRange
mc:scenarioMechanism	mc:Scenario	mc:MechanismSpecification
mc:scenarioPolicy	mc:Scenario	mc:Policy
mc:supportsPolicy	mc:MechanismSpecification	mc:Policy
mc:tracedToMechanism	mc:Requirement	mc:MechanismSpecification
mc:tracedToPolicy	mc:Requirement	mc:Policy
mc:usesAcceptanceProfile	mc:ValidationCampaign	mc:AcceptanceProfile
mc:usesIdentifier	mc:TransferEvent	mc:Identifier
mc:usesResolutionStrategy	mc:PolicyCategoryRelation	mc:ConflictResolutionStrategy
mc:usesStandard	mc:Artifact	mc:StandardSpecification
mc:validationUsesOntology	mc:ValidationCampaign	mc:OntologyArtifact
mc:validationUsesPolicies	mc:ValidationCampaign	mc:PolicyArtifact
mc:validationUsesQueries	mc:ValidationCampaign	mc:QueryCatalog
mc:validationUsesScenarios	mc:ValidationCampaign	mc:ScenarioArtifact

Datatype properties

Property	Asserted domain	Asserted range
mc:AHP_consistency_threshold	mc:AcceptanceProfile	xsd:decimal
mc:D_delegation_max	mc:AcceptanceProfile	xsd:integer
mc:E_device_max	mc:AcceptanceProfile	xsd:decimal
mc:N_agents	mc:AcceptanceProfile	xsd:integer
mc:T_decision_max	mc:AcceptanceProfile	xsd:decimal
mc:T_inference_local	mc:AcceptanceProfile	xsd:decimal
mc:T_migration_max	mc:AcceptanceProfile	xsd:decimal
mc:T_node_join	mc:AcceptanceProfile	xsd:decimal
mc:T_reselection_max	mc:AcceptanceProfile	xsd:decimal
mc:T_sparql_monitor	mc:AcceptanceProfile	xsd:decimal
mc:artifactIdentifier	mc:Artifact	xsd:string
mc:artifactStatus	mc:Artifact	xsd:string
mc:artifactVersion	mc:Artifact	xsd:string
mc:authorizationReason	mc:AuthorizationDecision	xsd:string
mc:categoryCode	mc:PolicyCategory	xsd:string
mc:conditionExpression	mc:RecoveryCondition	xsd:string
mc:configurationStatus	not declared	xsd:string
mc:containsPersistentIdentifier	mc:FederatedLearningSession	xsd:boolean
mc:containsPersonalData	mc:FederatedLearningSession	xsd:boolean
mc:hasPolicyAction	mc:Policy	xsd:string

Property	Asserted domain	Asserted range
mc:hasPolicyStatement	mc:Policy	rdfs:Literal
mc:identifierScope	mc:Identifier	xsd:string
mc:identifierValue	mc:Identifier	xsd:string
mc:mechanismDescription	mc:MechanismSpecification	rdfs:Literal
mc:mechanismIdentifier	mc:MechanismSpecification	xsd:string
mc:policyIdentifier	mc:Policy	xsd:string
mc:policyVersion	mc:Policy	xsd:string
mc:protectsAtRest	mc:SecurityMechanism	xsd:boolean
mc:protectsInTransit	mc:SecurityMechanism	xsd:boolean
mc:queryIdentifier	mc:QuerySpecification	xsd:string
mc:queryInterpretation	mc:QuerySpecification	xsd:string
mc:queryPurpose	mc:QuerySpecification	xsd:string
mc:relationCode	mc:PolicyRelationType	xsd:string
mc:requirementIdentifier	mc:Requirement	xsd:string
mc:requirementStatement	mc:Requirement	xsd:string
mc:scenarioIdentifier	mc:Scenario	xsd:string
mc:securityBaselineVersion	mc:SecurityMechanism	xsd:string
mc:strategyCode	mc:ConflictResolutionStrategy	xsd:string

Orchestration

21 named classes | 78 properties

Classes and asserted parents

Class	subClassOf	English label
mc:AdaptabilityLevel	-	Adaptability Level
mc:AdaptationAction	mc:TemporalEntity	Adaptation Action
mc:BufferRecord	mc:TemporalEntity	Buffer Record
mc:DecisionAlternative	-	Decision Alternative
mc:DecisionMethod	-	Decision Method
mc:DegradationEvent	mc:AdaptationAction	Degradation Event
mc:DelegationEvent	mc:AdaptationAction mc:TemporalEntity	Delegation Event
mc:MigrationCostLevel	-	Migration Cost Level
mc:MigrationEvent	mc:AdaptationAction	Migration Event
mc:MigrationTimeLevel	-	Migration Time Level
mc:ModelDegradationCause	-	Model Degradation Cause
mc:ModelSelectionAction	mc:AdaptationAction	Model Selection Action
mc:OffloadingEvent	mc:AdaptationAction	Offloading Event
mc:PairwiseComparison	-	Pairwise Comparison
mc:ParametrizedData	-	Parametrized Data
mc:ProfitabilityLevel	-	Profitability Level
mc:ReplicationEvent	mc:TemporalEntity	Replication Event
mc:RetentionEvent	mc:AdaptationAction	Retention Event
mc:RollbackEvent	mc:AdaptationAction	Rollback Event
mc:ScalingEvent	mc:AdaptationAction	Scaling Event
mc:SynchronizationEvent	mc:AdaptationAction	Synchronization Event

Object properties

Property	Asserted domain	Asserted range
mc:actionOriginNode	mc:AdaptationAction	mc:ComputationalNode
mc:actionTargetNode	mc:AdaptationAction	mc:ComputationalNode
mc:affectsModel	mc:AdaptationAction	mc:AIModel
mc:affectsService	mc:AdaptationAction	mc:Service
mc:alternativeModelTier	mc:DecisionAlternative	mc:ModelTier
mc:appliesPolicy	mc:EvaluationState	mc:Policy
mc:auditsContract	mc:EvaluationState	mc:SemanticContract
mc:authorizedByEvaluation	mc:AdaptationAction	mc:EvaluationState
mc:carriesData	mc:TransferEvent	not declared
mc:criterionA	mc:PairwiseComparison	not declared
mc:criterionB	mc:PairwiseComparison	not declared
mc:delegatedBy	mc:DelegationEvent	mc:ComputationalNode
mc:delegatesTo	(mc:ComputationalNode OR mc:DelegationEvent)	mc:ComputationalNode
mc:derivedFrom	mc:State	sosa:Observation

Property	Asserted domain	Asserted range
mc:evaluatesNode	mc:EvaluationState	mc:ComputationalNode
mc:evaluatesService	mc:EvaluationState	mc:Service
mc:evaluationPurpose	mc:EvaluationState	mc:ProcessingPurpose
mc:evaluationUser	mc:EvaluationState	mc:User
mc:evaluationZone	mc:EvaluationState	mc:CityArea
mc:expiresOnRecoveryOf	mc:DelegationEvent	mc:ComputationalNode
mc:hasAdaptability	mc:ServiceState	mc:AdaptabilityLevel
mc:hasDecisionAlternative	mc:EvaluationState	mc:DecisionAlternative
mc:hasDecisionMethod	mc:EvaluationState	mc:DecisionMethod
mc:hasDegradationCause	not declared	mc:ModelDegradationCause
mc:hasDelegation	mc:ComputationalNode	mc:DelegationEvent
mc:hasDetectedSymptom	mc:EvaluationState	mc:MAPESymptom
mc:hasMigrationCost	mc:EvaluationState	mc:MigrationCostLevel
mc:hasMigrationTime	mc:EvaluationState	mc:MigrationTimeLevel
mc:hasModelTier	mc:AIModel	mc:ModelTier
mc:hasPairwiseComparison	mc:EvaluationState	mc:PairwiseComparison
mc:hasProcessingLevel	mc:ServiceState	mc:ProcessingLevel
mc:hasProfitability	mc:EvaluationState	mc:ProfitabilityLevel
mc:hasRecoveryCondition	mc:DelegationEvent	mc:RecoveryCondition
mc:hasReplicationEvent	not declared	mc:ReplicationEvent
mc:hasServiceState	mc:Service	mc:ServiceState
mc:hostedOnNode	mc:AIModel	mc:ComputationalNode
mc:hostsModel	mc:ComputationalNode	mc:AIModel
mc:hostsService	mc:ComputationalNode	mc:Service
mc:parentDelegation	mc:DelegationEvent	mc:DelegationEvent
mc:partOfScenario	not declared	mc:Scenario
mc:replicationSource	mc:ReplicationEvent	not declared
mc:replicationTarget	mc:ReplicationEvent	not declared
mc:resultedInAction	mc:EvaluationState	mc:AdaptationAction
mc:rollbackTarget	mc:RollbackEvent	mc:AIModel
mc:selectedAlternative	mc:EvaluationState	mc:DecisionAlternative
mc:selectedModelTier	mc:EvaluationState	mc:ModelTier
mc:supersedesModel	mc:AIModel	mc:AIModel
mc:transferDestination	mc:TransferEvent	not declared
mc:transferSource	mc:TransferEvent	not declared
mc:triggeredByState	(mc:DelegationEvent OR mc:EvaluationState)	mc:State
mc:usesModel	mc:Service	mc:AIModel

Datatype properties

Property	Asserted domain	Asserted range
mc:delegationDepth	mc:DelegationEvent	xsd:integer
mc:eligibilityReason	mc:DecisionAlternative	xsd:string
mc:estimatedPredictionError	mc:EvaluationState	xsd:decimal
mc:evaluationOrder	mc:PolicyCategory	xsd:integer

Property	Asserted domain	Asserted range
mc:hasAHPScore	mc:DecisionAlternative	xsd:decimal
mc:hasConsistencyRatio	mc:EvaluationState	xsd:decimal
mc:hasConsistencyThreshold	mc:EvaluationState	xsd:decimal
mc:hasEvaluationTicketID	mc:EvaluationState	xsd:string
mc:hasLatencyWeight	mc:EvaluationState	xsd:decimal
mc:hasModelQualityWeight	mc:EvaluationState	xsd:decimal
mc:hasPrivacyWeight	mc:EvaluationState	xsd:decimal
mc:hasSelectionJustification	mc:EvaluationState	xsd:string
mc:hasTrustWeight	mc:EvaluationState	xsd:decimal
mc:idempotencyKey	not declared	xsd:string
mc:isEligible	mc:DecisionAlternative	xsd:boolean
mc:lastUpdated	mc:AIModel	xsd:dateTime
mc:legacyDecisionScore	mc:EvaluationState	xsd:decimal
mc:migrationNote	not declared	xsd:string
mc:migrationStatus	not declared	xsd:string
mc:modelLineageStatus	mc:AIModel	xsd:string
mc:modelVersion	mc:AIModel	xsd:string
mc:observedModelQuality	mc:EvaluationState	xsd:decimal
mc:pairwiseValue	mc:PairwiseComparison	xsd:decimal
mc:plannedExpiry	mc:TemporalEntity	xsd:dateTime
mc:predictionConfidence	mc:EvaluationState	xsd:decimal
mc:replicationVersion	not declared	xsd:string
mc:requiresRecalculation	mc:EvaluationState	xsd:boolean

Federated learning

2 named classes | 22 properties

Classes and asserted parents

Class	subClassOf	English label
mc:ModelGradientUpdate	mc:ParametrizedData	Model Gradient Update
mc:PrivacyBudgetAccount	-	Privacy Budget Account

Object properties

Property	Asserted domain	Asserted range
mc:aggregatesData	mc:ComputationalNode	mc:ParametrizedData
mc:budgetForContract	mc:PrivacyBudgetAccount	mc:SemanticContract
mc:budgetForPurpose	mc:PrivacyBudgetAccount	mc:ProcessingPurpose
mc:hasPayloadType	mc:FederatedLearningSession	mc:PayloadType
mc:hasPrivacyBudgetAccount	mc:FederatedLearningSession	mc:PrivacyBudgetAccount
mc:hasPrivacyMechanism	(mc:FederatedLearningSession OR mc:ParametrizedData)	mc:PrivacyMechanism
mc:involvedNode	mc:FederatedLearningSession	mc:ComputationalNode
mc:originatesFromDevice	mc:ParametrizedData	mc:MobileDevice
mc:sentToNode	mc:ParametrizedData	mc:ComputationalNode
mc:transfersData	mc:FederatedLearningSession	mc:ParametrizedData
mc:updatesModel	mc:FederatedLearningSession	mc:AIModel

Datatype properties

Property	Asserted domain	Asserted range
mc:containsIndividualizedGradients	mc:FederatedLearningSession	xsd:boolean
mc:hasAnonymizationApplied	mc:ParametrizedData	xsd:boolean
mc:hasNoiseApplied	mc:ParametrizedData	xsd:boolean
mc:hasPrivacyBudget	mc:FederatedLearningSession	xsd:decimal
mc:isRedundant	mc:ParametrizedData	xsd:boolean
mc:noiseLevel	mc:FederatedLearningSession	xsd:decimal
mc:privacyBudgetConsumed	mc:PrivacyBudgetAccount	xsd:decimal
mc:privacyBudgetMaximum	mc:PrivacyBudgetAccount	xsd:decimal
mc:privacyBudgetRemaining	mc:PrivacyBudgetAccount	xsd:decimal
mc:sendTimestamp	mc:ParametrizedData	xsd:dateTime
mc:sessionTime	mc:FederatedLearningSession	xsd:dateTime

Deployment

4 named classes | 7 properties

Classes and asserted parents

Class	subClassOf	English label
mc:AuthorityScope	-	Authority scope
mc:DeploymentFragment	-	Deployment fragment
mc:PrivacyClass	-	Privacy class
mc:SemanticNodeProfile	-	Semantic node profile

Object properties

Property	Asserted domain	Asserted range
mc:authoritativeFor	mc:DeploymentFragment	mc:AuthorityScope
mc:hasPrivacyClass	mc:DeploymentFragment	mc:PrivacyClass
mc:hostedAt	mc:DeploymentFragment	mc:ComputationalNode
mc:replicaOf	mc:DeploymentFragment	mc:DeploymentFragment

Datatype properties

Property	Asserted domain	Asserted range
mc:graphIRI	mc:DeploymentFragment	xsd:anyURI
mc:ontologyDigest	mc:SemanticNodeProfile	xsd:string
mc:usesEntailmentProfile	mc:SemanticNodeProfile	xsd:string

Wellbeing (optional)

19 named classes | 7 properties

Classes and asserted parents

Class	subClassOf	English label
mc:AccelerometerSensor	mc:PhysiologicalSensor	Accelerometer Sensor
mc:EDASensor	mc:PhysiologicalSensor	EDA Sensor
mc:HeartRateSensor	mc:PhysiologicalSensor	Heart Rate Sensor
mc:MeasurementUnit	-	Measurement unit
mc:PersonStatus	-	Person Status
mc:PhysiologicalObservation	sosa:Observation	Physiological Observation
mc:PhysiologicalParametrizedData	mc:ParametrizedData	Physiological Parametrized Data
mc:PhysiologicalSensor	sosa:Sensor	Physiological Sensor
mc:SleepObservation	sosa:Observation	Sleep Observation
mc:SleepParametrizedData	mc:ParametrizedData	Sleep Parametrized Data
mc:SleepSensor	mc:PhysiologicalSensor	Sleep Sensor
mc:SmartBand	mc:Wearable	Smart Band
mc:SmartRing	mc:Wearable	Smart Ring
mc:SmartWatch	mc:Wearable	Smart Watch
mc:SpO2Sensor	mc:PhysiologicalSensor	SpO2 Sensor
mc:StressLevel	-	Stress Level
mc:StressObservation	sosa:Observation	Stress Observation
mc:TemperatureSensor	mc:PhysiologicalSensor	Skin Temperature Sensor
mc:Wearable	saref:Device	Wearable

Object properties

Property	Asserted domain	Asserted range
mc:generatedBy	mc:ParametrizedData	mc:Wearable
mc:hasDeviceState	mc:Wearable	mc:DeviceState
mc:hasMeasurementUnit	sosa:Observation	mc:MeasurementUnit
mc:hasPersonStatus	mc:UserState	mc:PersonStatus
mc:hasPredictedStressLevel	mc:UserState	mc:StressLevel
mc:hasWearable	mc:User	mc:Wearable

Datatype properties

Property	Asserted domain	Asserted range
mc:measurementUnitStatus	sosa:Observation	xsd:string

External standards / SHACL

9 named classes | 23 properties

Classes and asserted parents

Class	subClassOf	English label
foaf:Person	-	Person
geo:Feature	-	Feature
geo:Geometry	-	Geometry
saref:Device	-	Device
sh:NodeShape	-	Node Shape
sosa:ObservableProperty	-	Observable Property
sosa:Observation	-	Observation
sosa:Platform	-	Platform
sosa:Sensor	-	Sensor

Object properties

Property	Asserted domain	Asserted range
geo:hasGeometry	not declared	not declared
sosa:hasFeatureOfInterest	not declared	not declared
sosa:isHostedBy	not declared	not declared
sosa:madeBySensor	not declared	not declared
sosa:observedProperty	not declared	not declared
sosa:observes	sosa:Sensor	sosa:ObservableProperty

Datatype properties

Property	Asserted domain	Asserted range
foaf:name	not declared	not declared
sosa:hasSimpleResult	not declared	not declared
sosa:resultTime	not declared	not declared

Annotation properties

Property	Asserted domain	Asserted range
dcterms:identifier	not declared	not declared
dcterms:language	not declared	not declared
dcterms:modified	not declared	not declared
dcterms:replaces	not declared	not declared
sh:class	not declared	not declared
sh:datatype	not declared	not declared
sh:maxCount	not declared	not declared
sh:message	not declared	not declared
sh:minCount	not declared	not declared
sh:or	not declared	not declared
sh:path	not declared	not declared
sh:property	not declared	not declared

Property	Asserted domain	Asserted range
sh:severity	not declared	not declared
sh:targetClass	not declared	not declared

Additional axioms and provenance

Subject	Axiom	Object
mc:DirectIdentifier	disjoint	mc:AnonymousIdentifier
mc:DirectIdentifier	disjoint	mc:PseudonymousIdentifier
mc:PseudonymousIdentifier	disjoint	mc:AnonymousIdentifier
mc:hasGeometry	subproperty	geo:hasGeometry

Full IRIs, expression members, declaration source files and comments are retained in data/schema.json and the CSV inventories. GraphML retains term identity and schema edges for graph editors, not the full OWL model. No additional ontology is imported from the network. The hashes below identify the exact 15 runtime inputs, including the deployment extension.

Source	SHA-256
ontology/core/schema.ttl	1f9111a1be516812dee7033721ac3f1a8d6cddc0667064ebf59065e7d1bddd34
ontology/modules/foundation.ttl	e461b109eb476d29aba978cbbbfb299f77fd8e795d7ac4b2b2ee6312f608a32f
ontology/modules/contracts.ttl	ddbf564e174057b042ad9bcf71691d7201f5dcd438cf7392bacecbe9564b3311
ontology/modules/topology.ttl	d0bb23abff836f066a6ada0849b0d0ee46ed67d17052f0f7e1b96ae5de2d04f2
ontology/modules/observability.ttl	269680d6465eec074eb57704c5f9dc247ed32396dee92dfc30b09e113331b4f6
ontology/modules/governance.ttl	82f348b264b76eec1da82ba36fa69c245390c77f53cc819a26e76f9c153907d4
ontology/modules/orchestration.ttl	84ebe567f6ae603d71a286d13eede9e98c6b23ca7f7d84162c5ad5940ee6f77c
ontology/modules/federation.ttl	79c87459e23e7e1bc748c823a5cc37d389b1fd5167dc41454574fe520fdaf044
ontology/modules/deployment.ttl	dc351bab13628cc8b0dbf25aba02bf3e0a34682a616ca8a863633fd5c60fdb60
ontology/core/vocabulary.ttl	455cf768208a275fc34e7335b7cd86751c91de110beb044871b2dab5092d803f
ontology/domains/wellbeing/schema.ttl	78dbf2815e1cd929fce693161dd3a5fb9dad9d287a157ad6c451ec2a4c26c38a
ontology/domains/wellbeing/vocabulary.ttl	b72a7fcc1e8062096a4b23c044caee9b9d97f823351f1dc4d38d042b962c2c94
ontology/examples/reference-system.ttl	4cefc7e5137a8faac5ad5613d7785a866292a93b923f8e5dec9a069d3d664732
ontology/shapes/core-compliance.ttl	b333eff8d2965d62d07c19785f6200f278507eca6fb4c5edf01efdb3c613bed5
ontology/shapes/federated-learning.ttl	4497eee9b46dd93dc5caf9e17a64f6c4cd8d55b1a7a8b2aeabeea91595674378